

PhishAware

Phishing Awareness Simulator Web Application

Cybersecurity Awareness & Educational Project Report

Project Type: Cybersecurity Awareness Web Application

Technology Stack: Flask, HTML, CSS, JavaScript

Purpose: Educational Phishing Awareness Simulation

1. Introduction

PhishAware is a cybersecurity awareness simulator developed for educational purposes. The project helps users understand how phishing attacks work and teaches them how to identify suspicious websites, emails, and credential harvesting attempts.

The platform simulates realistic phishing scenarios such as fake bank login pages, social media phishing pages, and giveaway scams. Instead of collecting credentials, the application redirects users to educational awareness pages explaining the phishing indicators and security risks.

2. Objective

The primary objective of PhishAware is to educate users about phishing attacks through interactive simulations and cybersecurity awareness training.

Objectives:

- Teach users how phishing attacks work
- Help users identify suspicious links and fake login pages
- Provide cybersecurity best practices
- Improve awareness against social engineering attacks
- Create an engaging learning environment using simulations and quizzes

3. Features

- Interactive phishing simulation pages
- Fake bank login demo
- Fake social media phishing page
- Fake giveaway scam simulation
- Cybersecurity awareness explanations
- Interactive phishing awareness quiz
- Progress tracking system
- Certificate generation system
- Security tips and best practices section
- URL safety checker
- Email analyzer
- Responsive dark cybersecurity-themed UI

4. Technologies Used

- Python Flask – Backend Framework
- HTML5 – Frontend Structure
- CSS3 – Styling & UI Design
- JavaScript – Interactive Features

- JSON – Quiz Data Storage
- Render – Deployment Platform
- Git & GitHub – Version Control

5. Project Workflow

User Visits Site



Chooses Simulation



Interacts With Fake Phishing Page



Awareness Explanation Displayed



Cybersecurity Tips Provided



Interactive Quiz



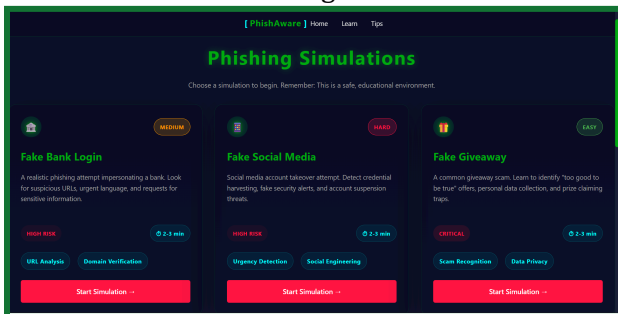
Certificate Generation

6. Screenshots

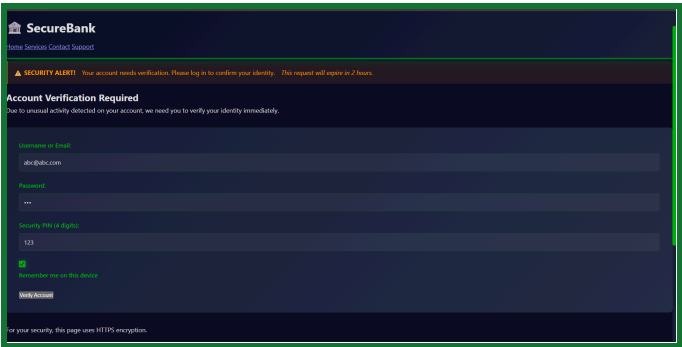
- Landing Page



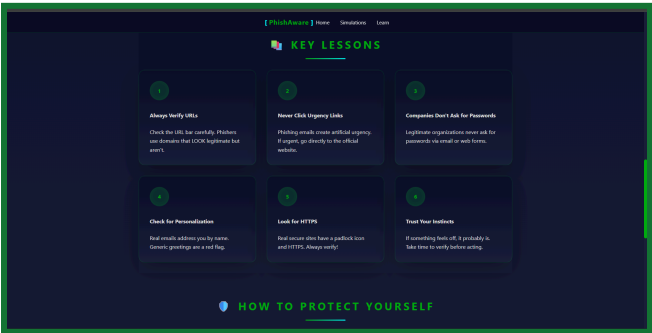
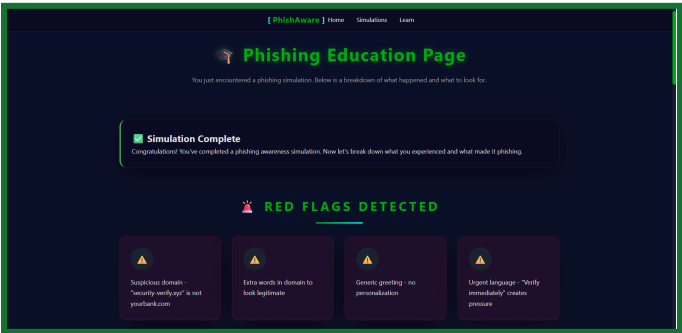
- Simulation Selection Page



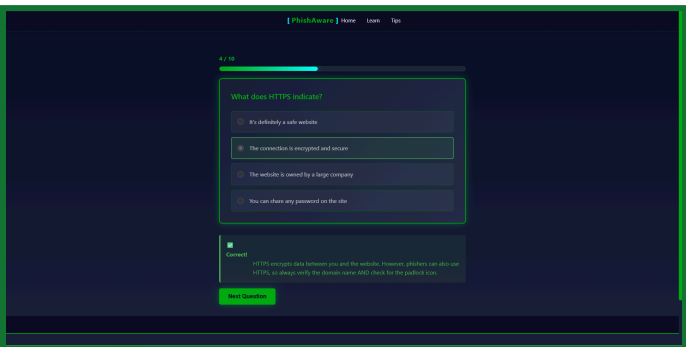
• Fake Bank Login Page



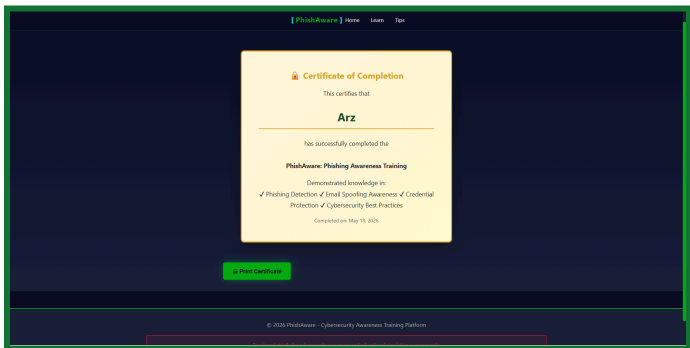
• Awareness Explanation Page



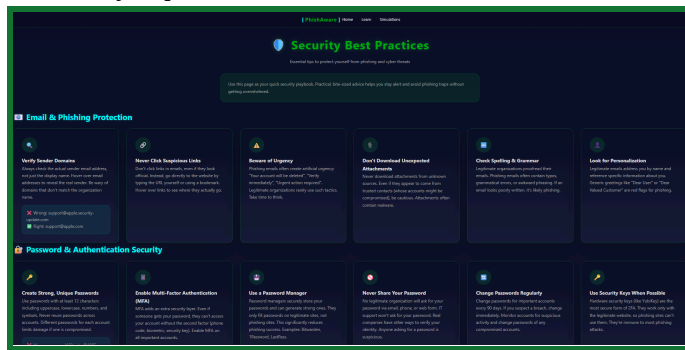
• Quiz Section



• Certificate Generation



- Security Tips Dashboard



7. Security Awareness Concepts

- Phishing
- Spear Phishing
- Email Spoofing
- Smishing
- Vishing
- Credential Harvesting
- Social Engineering
- Multi-Factor Authentication (MFA)

8. Challenges Faced

- Managing Flask and JavaScript integration
- Handling JSON quiz data properly
- Debugging frontend event listeners
- Creating responsive cybersecurity-themed UI
- Building dynamic progress tracking
- Generating printable certificates
- Implementing secure educational simulations

9. Learning Outcomes

- Flask backend development
- Frontend UI/UX design
- DOM manipulation with JavaScript
- Cybersecurity awareness concepts
- Phishing attack simulation
- Interactive web application architecture
- Debugging and troubleshooting
- Deployment and GitHub workflow

10. GitHub Link

<https://github.com/arzkhudia09-jpg/PhishAware.git>

11. Live Deployment Link

<https://phishaware-2dwn.onrender.com>

12. Conclusion

PhishAware successfully demonstrates how phishing attacks operate while educating users about cybersecurity awareness and prevention techniques.

The project combines interactive simulations, quizzes, and educational explanations to create a professional phishing awareness learning platform. It also highlights the importance of cybersecurity education in preventing credential theft and social engineering attacks.

This project strengthened practical web development skills as well as cybersecurity knowledge related to phishing detection and user awareness training.

13. Ethical Disclaimer

PhishAware is developed strictly for cybersecurity awareness, educational training, and phishing simulation purposes only.

No credentials are collected, stored, or transmitted. The project is intended to promote safe cybersecurity practices and user awareness.